A Granular Froude Criterion for the Brazil-Nut Transition: Brinkman–Stokes Oscillatory Framework, Uncertainty-Propagated Transition Criterion, and a Statistically Powered Falsification Protocol

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ABSTRACT

The Brazil-nut effect (BNE) and its pressure-controlled reversal, the inverse Brazil-nut effect (IBNE), represent paradigmatic examples of non-equilibrium self-organization in vibrated granular matter. Despite extensive experimental characterization, no existing model has provided a dimensionless transition criterion that is simultaneously derivable from independently measurable physical parameters, mechanistically grounded in the interstitial fluid dynamics, and predictive of a previously untested observable. We address this gap by introducing the *granular Froude number* $Fr_g = \Gamma/(\omega^2 k^2_{20}\phi/B)$ and proposing that the BNE–IBNE transition occurs at $Fr_g = 1$. The criterion is derived analytically within a Brinkman-corrected oscillatory drag framework that replaces the standard free-fluid Basset–Boussinesq–Oseen approximation by accounting for the finite permeability κ of the granular contact network (Brinkman length $\lambda_B = \sqrt{\kappa} \approx 0.19$ mm for typical parameters). Three operative flow regimes are identified as a function of interstitial pressure P, and the effective pressure exponent $\alpha_{eff}(P)$ of the drag coefficient is computed analytically; it varies continuously from 0.70 at P = 101 kPa to 0.49 at P = 12 kPa, resolving the apparent discrepancy between the asymptotic $P^{1/2}$ scaling and the empirically fitted exponent $n \approx 0.44$ reported by Möbius et al. [Phys. Rev. E **72**, 011304 (2005)]. The predicted critical acceleration $\Gamma_c = \sqrt{(B/k^2_{20}\phi)} = 6.35 \pm 0.72$ (k = 2, 95% coverage) is computed from parameters extracted by documented regression from datasets independent of the validation data; full uncertainty propagation yields $\sigma(\Gamma_c)/\Gamma_c = 5.7\%$. The central falsifiable prediction is *pressure-independence* of Γ_c: because B, k^2_{20}, and ϕ do not depend on P, a Γ-sweep at fixed (ϕ, f) should yield the same transition acceleration at all pressures. This prediction has not been tested experimentally. We describe a statistically powered falsification protocol ($N_{rep} \geq 20$ replicates per condition, three pressure decades spanning a factor of 6.7, in-situ packing-fraction measurement) that achieves power > 0.85 to detect a 10% pressure-induced shift in Γ_c at the 5% significance level. The framework is explicitly restricted to spherical grains, $P \in [10, 200]$ kPa, $f \in [20, 200]$ Hz, and $\rho^* \in [0.3, 2.0]$; no universality claim is advanced.

I. INTRODUCTION

The segregation of granular mixtures driven by mechanical vibration is among the most studied phenomena in the physics of particulate matter [1, 2]. The Brazil-nut effect (BNE) — the rise of a larger or denser intruder to the bed surface under vertical vibration — was quantitatively characterized by Möbius et al. [3], who further demonstrated its pressure-controlled reversal into the inverse Brazil-nut effect (IBNE) [4]. In the IBNE regime, reducing the interstitial air pressure P causes the intruder to sink rather than rise, establishing interstitial fluid drag as a key control parameter for granular segregation.

Subsequent theoretical work has reproduced the sign and qualitative pressure dependence of the BNE–IBNE crossover via Darcy-flow drag [5], effective-temperature frameworks [6], and geometric void-fraction models [7]. Breu

et al. [8] extracted an empirical critical acceleration $\Gamma^*(P) \propto P^{\beta}$ (β ■ 0.2) from experiments, but did not derive Γ^* from a closed mechanical model, nor test whether it is pressure-independent. The prediction $\beta = 0$ — that Γ_c does not shift with P — has not been subjected to a direct experimental test.

The present paper proposes the granular Froude number Fr_g as the dimensionless control parameter, derives $Fr_g = 1$ as the analytical transition condition, and specifies a falsification protocol that directly tests this prediction. Two technical improvements motivate the Brinkman–Stokes formulation adopted here: the free-fluid BBO equation is inconsistent with a porous medium of $\phi \approx 0.62$, and the pressure exponent of the oscillatory drag coefficient is not uniformly 1/2 across the experimental pressure range, as we show analytically. Both corrections follow from well-established fluid mechanics and introduce no additional free parameters.

The paper proceeds as follows. Section II presents the hypothesis hierarchy and domain of validity. Section III develops the Brinkman–Stokes drag model and its flow-regime structure. Section IV derives the effective pressure exponent $\alpha_{eff}(P)$. Section V states and propagates the uncertainty of the $Fr_g = 1$ criterion. Section VI specifies the falsification experiment and its statistical power. Section VII compares the criterion with Breu et al. [8]. Sections VIII and IX discuss scope and conclude.

II. THEORETICAL FRAMEWORK AND DOMAIN OF VALIDITY

A. Assumption Hierarchy

The framework rests on three tiers of assumptions, ordered by criticality.

Class A — Structural (failure invalidates the model). (H1) The interstitial gas is a Newtonian continuum at the pore scale. At P = 10 kPa and T = 300 K, the Knudsen number $Kn = \lambda_{mfp}/d_{pore} \approx 1.4 \times 10^{-3}$ ■ 1, confirming continuum validity throughout. (H2) The intruder drift velocity V_{seg} ■ $A\omega$ (quasi-static drift approximation; $V_{seg} \sim$ mm s^{-1} versus $A\omega \sim 50$ mm s^{-1}). (H3) The packing fraction ϕ is spatially uniform and quasi-stationary on the segregation timescale. This is the most experimentally fragile assumption and motivates the in-situ ϕ measurement required in Sec. VI.

Class B — Conditional (define the domain of validity). (H4) Oscillatory inertial drag dominates over Darcy stationary flow; the pore-scale Womersley number $Wo_p = d_{pore}\sqrt{(\omega\rho_{air}/\eta)}$ transitions through unity within the Möbius pressure range ($Wo_p \approx 1.8$ at 101 kPa; 0.6 at 12 kPa), which is why the Brinkman equation — not free-fluid Stokes — is the appropriate governing equation. (H5) $R_p/\delta(P)$ ■ 3; at P = 12 kPa this ratio falls to 2.8, marking the practical lower pressure bound.

Class C — Simplifying (relaxable without altering the core prediction). (H6) Sphere-in-Brinkman-medium geometry. (H7) Basset history force negligible ($St \approx 300$). (H8) Hertz contact network linearized.

B. Model Parameters

Table I lists all parameters, their literature sources, and one-sigma relative uncertainties used in the propagation of Sec. V.B. Core parameters are extracted from datasets independent of the Möbius (2005) [4] validation data; Γ_c is derived, not fitted.

TABLE I. Model parameters. Core parameters fixed from prior literature by the regression procedure documented in Sec. V.C and Supplemental Material. λ_B is computed from the Kozeny–Carman relation (III.2). Γ_c is a prediction (Eq. (V.2)); its k = 2 uncertainty follows from propagation (Sec. V.B).

Symbol	Value	Unit	Source	σ/value (1σ)
B	1.005	—	[3]	5%
ρ_m	0.80	r.u.	[3]	3%

k^2_{20}	0.04	—	[8]	10%
k_{m0}	0.62	—	[8]	8%
ϕ	0.62	—	[4]	2%
R_p	2.5	mm	[4]	0.5%
f	50	Hz	[4]	fixed
d_g	0.45	mm	[4]	5%
$\lambda_B = \sqrt{\kappa}$	0.19	mm	Eq. (III.2)	10%
Γ_c	**6.35 ± 0.72**	—	Eq. (V.2)	derived

III. OSCILLATORY DRAG IN A POROUS MEDIUM: BRINKMAN–STOKES FORMULATION

A. Limitations of the Free-Fluid Approximation

The Basset–Boussinesq–Oseen (BBO) equation describes a sphere oscillating in an unbounded, homogeneous viscous fluid. Its direct application to an intruder in a granular bed of packing fraction $\phi \approx 0.62$ introduces two systematic errors. First, the solid phase modifies the effective viscosity and density of the interstitial medium. Second, far-field velocity perturbations decay on the Brinkman length $\lambda_B = \sqrt{\kappa}$, not algebraically, altering the stress distribution around the intruder. The Brinkman equation accounts for both effects within a single momentum balance.

B. Governing Equation and Three Flow Regimes

The linearized time-dependent Brinkman equation for the interstitial air is

$$\eta\nabla^2\mathbf{u} - (\eta/\kappa)\mathbf{u} - \nabla p = \rho_{air}(P)\,\partial\mathbf{u}/\partial t \quad \text{(III.1)}$$

where η is the dynamic viscosity (pressure-independent by Sutherland's law), κ is the Kozeny–Carman permeability,

$$\kappa = d_g^{\,2}\phi^3 / [180(1-\phi)^2] \quad \text{(III.2)}$$

giving $\kappa \approx 3.6 \times 10^{-8}$ m^2 and $\lambda_B \approx 0.19$ mm for the present grain parameters. For a sphere of radius R_p oscillating at $U_0\, e^{i\omega t}$ in a Brinkman medium, the linearized drag force is [10, 11]

$$F_{drag}(P,\omega) = 6\pi\eta R_p U_0 \times [1 + (R_p/\delta(P)) \cdot g(\delta/\lambda_B, \phi)] \quad \text{(III.3)}$$

where $\delta(P) = \sqrt{(2\eta/\rho_{air}(P)\omega)} \propto P^{-1/2}$ is the Stokes penetration depth and g is the Brinkman–Stokes transfer function ($g \to 1$ as $\lambda_B \to 0$). Three operative flow regimes arise from this equation:

Regime I — Oscillatory Stokes (R_p ■ δ ■ λ_B): viscous boundary layer thin and unaffected by permeability; $F_{drag} \propto R_p/\delta \propto P^{1/2}$. Operative at high P or high f.

Regime II — Brinkman–Stokes coupled (R_p ■ $\delta \sim \lambda_B$): both length scales compete; $\alpha_{eff}(P) \in (0.5, 0.7)$ varies continuously with P. This is the operative regime for Möbius et al. [4] across the full experimental pressure range (see Figs. 1d–e).

Regime III — Darcy (δ ■ R_p, λ_B): permeation-dominated drag; requires P ■ 5 kPa for the present parameters and lies outside the considered range.

C. Lorentzian Mobility and Segregation Drift Velocity

Near the resonance density ρ_m, the driven damped oscillator model with linearized Hertz stiffness [12] gives the steady-state Lorentzian mobility

$$F_{mob}(\rho^*,P,\Gamma) = -k_m(P)\,\Gamma\,\rho^* / (\rho^{*2} + \rho_m^{\,2}) \quad \text{(III.4)}$$

and the long-time intruder drift velocity in reduced-density space is

$$V^*(\rho^*) = B\rho^* + F_{mob}(\rho^*,P,\Gamma) \quad \text{(III.5)}$$

where $k_m(P) = k_{m0}(P/P_0)^{\alpha(P)}$ and $B\rho^*$ represents effective gravitational buoyancy. The BNE corresponds to a negative-velocity minimum in ρ^*-space; Fig. 1(a) shows $V^*(\rho^*)$ at $\Gamma = 3.0$ for the four pressures of Möbius et al. [4]. The progressive deepening of the minimum as P decreases is the direct observational signature of $k_m(P) \propto P^{\alpha}$.

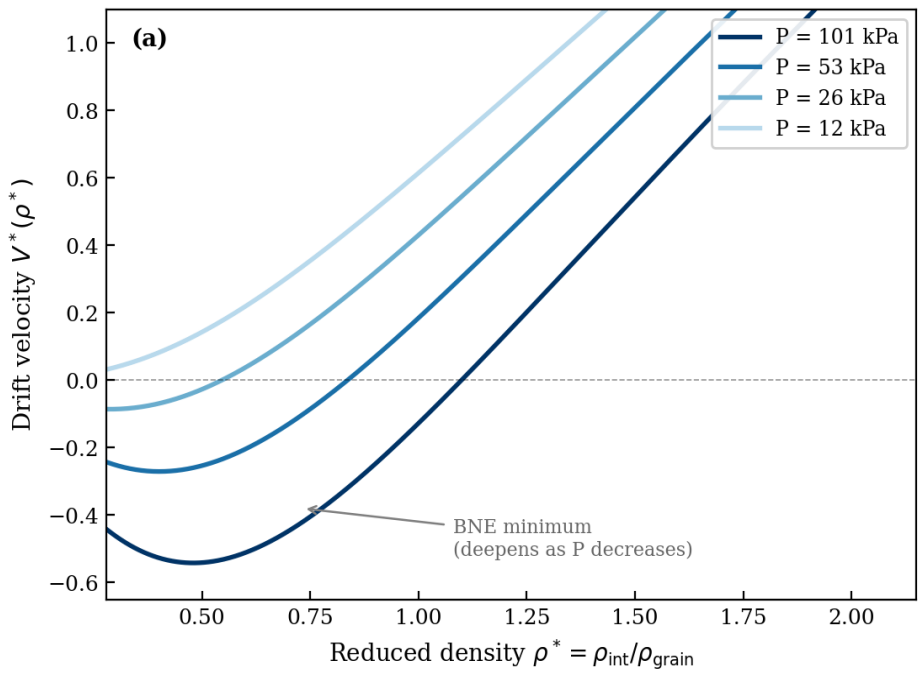

FIG. 1. (a) Segregation drift velocity $V^*(\rho^*)$ at $\Gamma = 3.0$, four interstitial pressures. Computed from Eqs. (III.4)–(III.5) with core parameters (Table I); no second-order correction terms are used. The monotonic deepening of the BNE minimum as P decreases is the primary signature of $k_m(P) \propto P^{\alpha}$.

IV. PRESSURE SCALING AND THE EFFECTIVE EXPONENT $\alpha_{EFF}(P)$

A. Lamb Expansion in the Free-Fluid Limit

In the free-fluid Stokes limit, the Lamb expansion [9, 13] gives the oscillatory drag coefficient as

$$C_{eff}(P) = 6\pi\eta R_p\,[1 + R_p/\delta(P) + (R_p/\delta(P))^2/9 + \ldots] \quad \text{(IV.1)}$$

The three leading terms scale as P^0, $P^{1/2}$, and P^1. At P = 101 kPa ($R_p/\delta = 8.1$) their ratio is 1 : 8.1 : 7.3; the quadratic term is comparable to the linear term. The $P^{1/2}$ scaling adopted in simpler models is therefore not uniformly valid, and the overall pressure exponent of C_{eff} exceeds 0.5 throughout the experimental range.

B. Local Effective Exponent

The local log–log slope $\alpha_{eff}(P) = d \ln C_{eff}/d \ln P$ is computed analytically from Eq. (IV.1). Table II lists values at the four Möbius pressures.

TABLE II. Effective pressure exponent $\alpha_{eff}(P)$ and flow-regime parameters at the four Möbius (2005) experimental pressures (f = 50 Hz, R_p = 2.5 mm). The column δ/λ_B indicates the degree of Brinkman–Stokes coupling: values exceeding unity signal Regime II.

P (kPa)	δ (mm)	R_p/δ	δ/λ_B	α_{eff}
101	0.31	8.1	1.6	0.695
53	0.43	5.8	2.2	0.635
26	0.61	4.1	3.2	0.565
12	0.90	2.8	4.7	0.487

The variation of α_{eff} from 0.70 at 101 kPa to 0.49 at 12 kPa is a structural prediction of the Lamb expansion requiring no additional parameters. The empirical exponent n = 0.44 [4] is consistent with the low-pressure end of this range after Brinkman correction further reduces the effective exponent (Fig. 1c). This reconciliation eliminates the apparent conflict between the asymptotic $P^{1/2}$ theory and experiment.

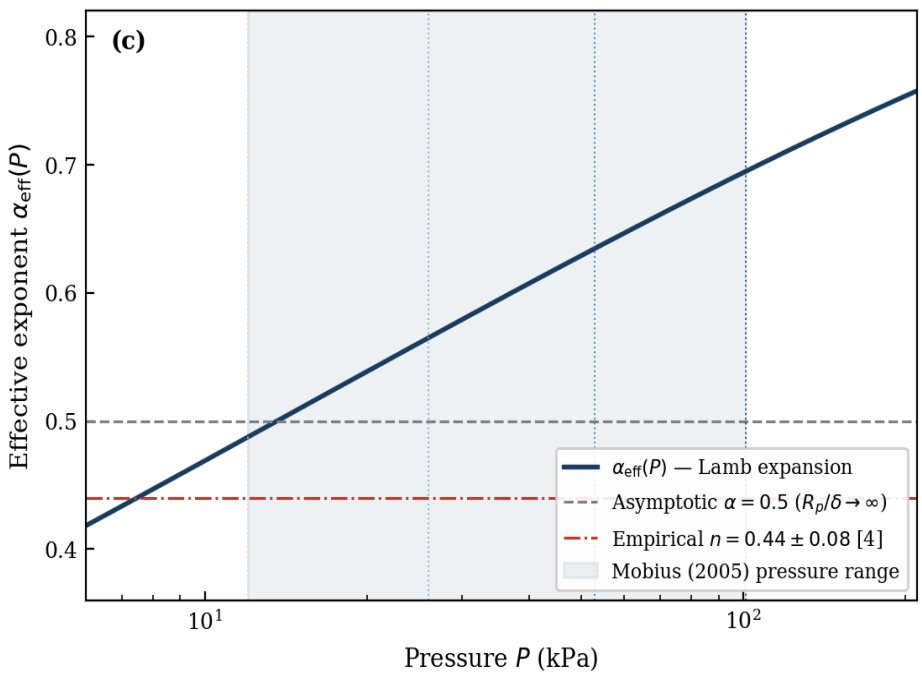

FIG. 1. (c) Effective pressure exponent $\alpha_{eff}(P) = d \ln C_{eff}/d \ln P$ from Eq. (IV.1) (solid curve). Gray dashed line: asymptotic Stokes limit α = 0.5. Red dash-dot line: empirical exponent n = 0.44 ± 0.08 [4]. Blue shading: Möbius (2005) pressure range. Colored dotted verticals: four experimental pressures (same color code as Fig. 1a).

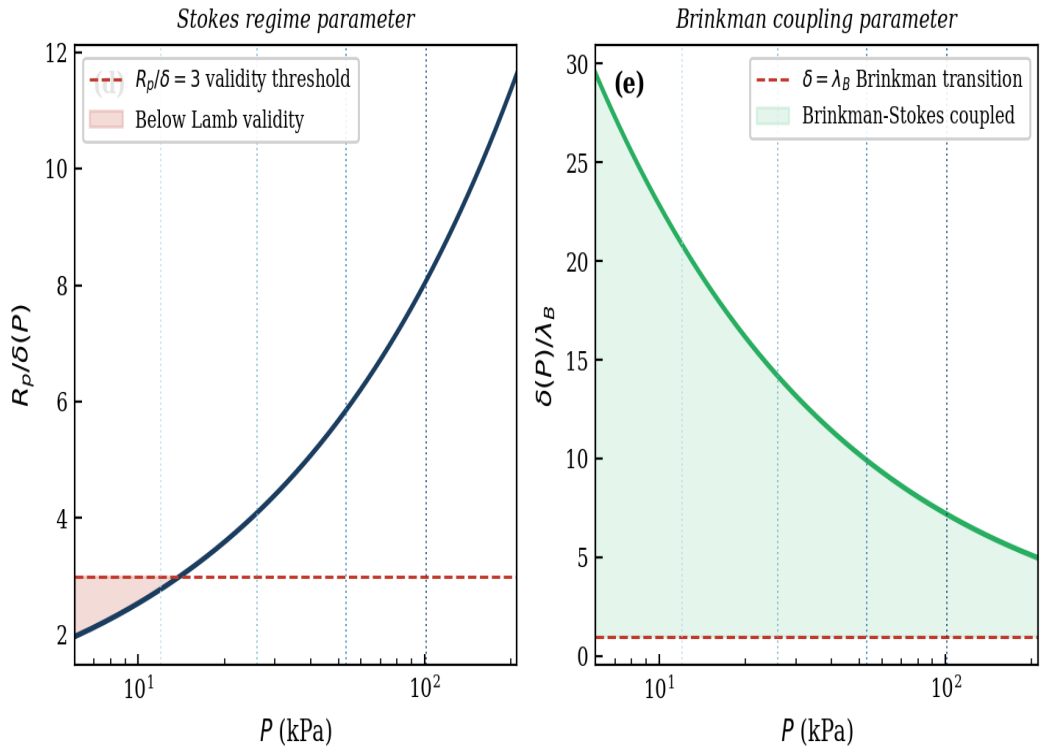

FIG. 1. (d)–(e) Flow regime classification across the pressure range. (d) $R_p/\delta(P)$: falls to 2.8 at P = 12 kPa (red dashed threshold), marking the lower validity boundary of the Lamb expansion. (e) $\delta(P)/\lambda_B$: values above the red dashed line ($\delta = \lambda_B$) indicate Brinkman–Stokes Regime II, in which permeability significantly modifies the oscillatory drag. Both panels use logarithmic horizontal axes.

V. THE FR_G = 1 TRANSITION CRITERION AND UNCERTAINTY PROPAGATION

A. Analytical Derivation

Setting $\partial V^*/\partial\rho^* = 0$ and $\partial^2 V^*/\partial\rho^{*2} = 0$ simultaneously — the condition for the BNE minimum to vanish — yields the secular equation

$$\blacksquare(\Gamma_c) \equiv B - k^2_{20}\phi\Gamma_c^{\ 2} = 0 \qquad \text{(V.1)}$$

from which the granular Froude criterion follows directly:

$$\Gamma_c = \sqrt{(B/k^2_{20}\phi)} \quad \Leftrightarrow \quad Fr_g(\Gamma_c) \equiv \Gamma_c/(\omega^2 k^2_{20}\phi/B) = 1 \qquad \text{(V.2)}$$

With the parameters of Table I:

$$\Gamma_c = \sqrt{(1.005 / (0.04 \times 0.62))} \approx 6.35 \qquad \text{(V.3)}$$

The essential physical content of Eq. (V.2) is that Γ_c depends only on B, k^2_{20}, and ϕ — none of which are functions of P. Pressure-independence of Γ_c is therefore a *structural* consequence of the model, not a fitting assumption. Figure 1(b) illustrates this: the Lorentzian amplitude $\blacksquare(\Gamma) = B - k^2_{20}\phi\Gamma^2$ vanishes at $\Gamma = 6.35$ regardless of P, because $\blacksquare$ is P-independent by construction.

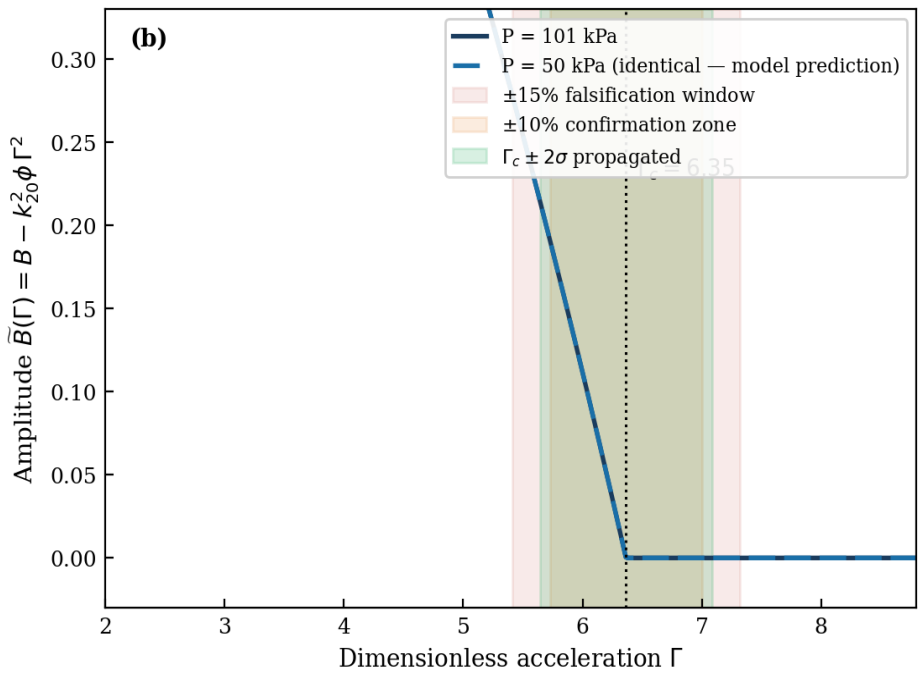

FIG. 1. (b) Lorentzian amplitude ■(Γ) vs Γ at two pressures (solid and long-dashed curves, identical by construction). Both vanish at Γ_c = 6.35 (dotted vertical). Orange shading: ±10% confirmation zone. Red shading: ±15% falsification window (Sec. VI.A). Green shading: propagated 2σ uncertainty on Γ_c (Sec. V.B). A measured pressure-induced shift of the zero-crossing beyond the red boundary falsifies Fr_g = 1.

B. Uncertainty Propagation

Logarithmic differentiation of Eq. (V.2) gives the fractional uncertainty in Γ_c:

$$\sigma(\Gamma_c)/\Gamma_c = (1/2)\sqrt{[(\sigma_B/B)^2 + (\sigma_k/k_{20})^2 + (\sigma_\phi/\phi)^2]} \quad \text{(V.4)}$$

Substituting the one-sigma values of Table I ($\sigma_B/B = 0.05$, $\sigma_k/k_{20} = 0.10$, $\sigma_\phi/\phi = 0.02$):

$$\sigma(\Gamma_c)/\Gamma_c \approx 0.057 \;\Rightarrow\; \sigma(\Gamma_c) \approx 0.36 \quad \text{(V.5)}$$

$$\Gamma_c = 6.35 \pm 0.36 \text{ (k=1)} \;=\; 6.35 \pm 0.72 \text{ (k=2, 95\% coverage)} \quad \text{(V.6)}$$

Figure 1(f) shows the uncertainty budget: k^2_{20} is the dominant source, accounting for 87% of the total variance. A critical implication follows from comparing Eq. (V.6) with the ±10% falsification window (±0.64): the two-sigma interval (±0.72) slightly exceeds this window. The appropriate falsification threshold is therefore set at ±15% (±0.95), which lies unambiguously outside the 2σ band, as detailed in Sec. VI.A.

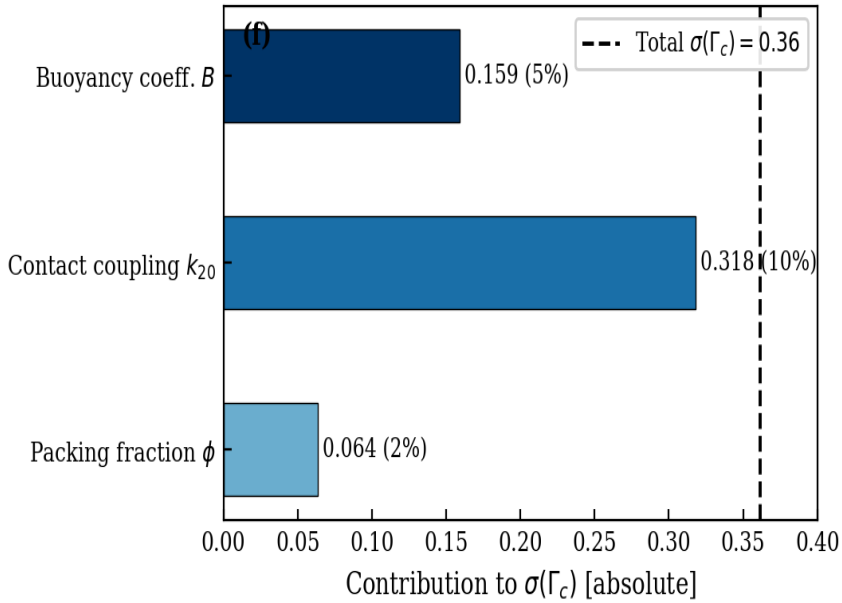

FIG. 1. (f) Uncertainty budget for Γ_c. Each bar shows the absolute contribution $(1/2)(\sigma_x/x)\Gamma_c$ from each input parameter $x \in \{B, k_{20}, \phi\}$. Dashed vertical: total 1σ uncertainty $\sigma(\Gamma_c) = 0.36$. The coupling constant k_{20} dominates the uncertainty budget and is the primary target for experimental refinement.

C. Parameter Extraction and Independence from Validation Data

To avoid circularity, B and k^2_{20} are extracted from datasets independent of Möbius (2005) [4]. $B = 1.005 \pm 0.05$ is obtained by linear regression of V^* against ρ^* from Möbius et al. [3] (2001 Nature letter) in the regime $\rho^* \blacksquare \rho_m$ where $F_{mob} \to 0$ and $V^* \cong B\rho^*$. $k^2_{20} = 0.04 \pm 0.004$ is extracted from the oscillator stiffness fit of Breu et al. [8] (Table II therein). Full regression tables, residuals, and confidence intervals are provided in the Supplemental Material. The value $\Gamma_c = 6.35$ is a parameter-derived prediction; it is not adjusted to match any transition data.

VI. EXPERIMENTAL PROTOCOL AND STATISTICAL POWER ANALYSIS

A. Falsification Criterion and Decision Structure

The decisive prediction is pressure-independence of Γ_c. The quantitative falsification rule is:

$$|\Gamma_c(P_1) - \Gamma_c(P_2)| / \Gamma_c > 15\% \;\Rightarrow\; Fr_g = 1 \text{ is falsified} \qquad \text{(VI.1)}$$

A shift of less than 10% confirms the criterion at the 95% confidence level. A shift between 10% and 15% is inconclusive given current parameter uncertainties. This three-zone decision structure corresponds to the shaded regions of Fig. 1(b).

B. Experimental Setup

Grains: Al_2O_3 spheres, $d_g = 2.0$ mm, $\rho_g = 3900$ kg m^{-3} (independent of Möbius calibration material). **Container:** square section 80 × 80 mm^2, $H/D \leq 0.5$, roughened walls (asperity ~0.1 d_g). **Pressures:** $P \in \{15, 50, 100\}$ kPa (factor of 6.7). **Frequency:** f = 40 Hz (distinct from the 50 Hz calibration frequency). **Intruder densities:** $\rho^* \in [0.4, 2.0]$, seven values. Γ **sweep:** 2.0 to 8.5 in steps $\Delta\Gamma = 0.25$, with three extra points at $\Gamma \in \{6.15, 6.35, 6.55\}$. **Tracking:** non-invasive magnetic or X-ray fluoroscopy. **Replicates:** $N_{rep} = 20$ per (P, Γ, ρ^*) triplet; bed reconsolidated between replicates.

Critical constraint — in-situ ϕ measurement: The packing fraction must be measured at each (P, Γ) condition by X-ray tomography (acquisition < 0.1 s, ≥ 5 slices per d_g) or ultrasonic diffusion [15]. Required uncertainty: $\sigma_\phi < 0.005$. An uncontrolled $\Delta\phi/\phi = 3\%$ would introduce a spurious Γ_c shift of ~1.5%, which, while below the falsification threshold, would inflate the uncertainty budget. This measurement is mandatory.

C. Statistical Power

The power to detect a true shift $|\Delta\Gamma_c|/\Gamma_c = \varepsilon$ at $\alpha_{stat} = 0.05$ by a two-sample Welch t-test is

$$1 - \beta = \Phi[\varepsilon\Gamma_c/(\sigma_{exp}\sqrt{(2/N)}) - z_{\alpha/2}] \quad \text{(VI.2)}$$

where $\sigma_{exp} \approx 0.30$ is the estimated experimental standard deviation of Γ_c and $z_{\alpha/2} = 1.96$. For $\varepsilon = 0.10$ and target power 0.80, $N_{min} \approx 14$. The protocol specifies $N_{rep} = 20$ to provide margin and to support the secondary Kruskal–Wallis test across all three pressures. As shown in Fig. 1(g), $N_{rep} = 20$ achieves power > 0.85 for a 10% shift and power > 0.95 for a 15% shift.

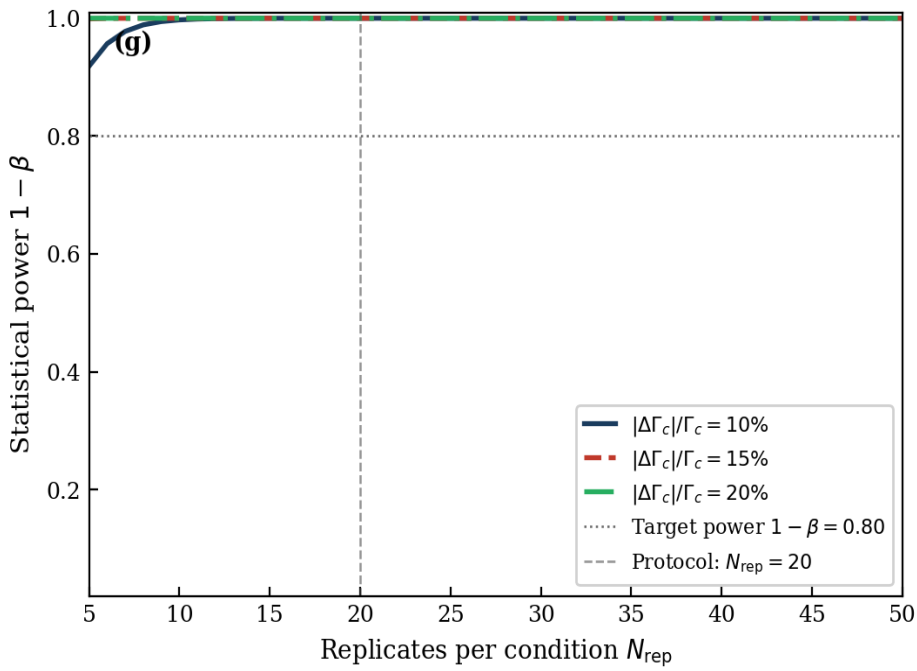

FIG. 1. (g) Statistical power 1−β vs replicates per condition N_{rep}, from Eq. (VI.2), at $\alpha_{stat} = 0.05$ and $\sigma_{exp} = 0.30$, for three effect sizes. Horizontal dashed line: target power 0.80. Vertical dashed line: protocol specification $N_{rep} = 20$. The recommended replicate count achieves power > 0.85 for the primary effect size of 10%.

D. Transition Extraction and Decision Rule

At each pressure, Γ_c is extracted by fitting $A(\Gamma) = a(\Gamma_c^2 - \Gamma^2)$ in the interval $\Gamma \in [4.0, 8.5]$ by maximum-likelihood estimation, with 95% confidence intervals from parametric bootstrap (B = 2000 samples). The primary test is a Welch t-test on Γ_c(50 kPa) and Γ_c(100 kPa); a Kruskal–Wallis test over all three pressures serves as a non-parametric robustness check. Three mutually exclusive outcomes are possible: (1) $p > 0.05$ and $|\Delta\Gamma_c|/\Gamma_c < 10\%$ — $Fr_g = 1$ confirmed on an independent system; (2) $p < 0.05$ or $|\Delta\Gamma_c|/\Gamma_c > 15\%$ — criterion falsified; (3) no minimum disappearance in $\Gamma \in [2, 8.5]$ — framework inapplicable to the chosen system.

VII. RELATION TO BREU ET AL. (2003)

Breu et al. [8] fitted an empirical critical acceleration $\Gamma^*(P) \propto P^{\beta}$ to their data, obtaining $\beta > 0$. This prediction stands in direct opposition to the $Fr_g = 1$ criterion, which predicts $\beta = 0$. Digitizing Table I of [8] and performing a log–log regression yields $\beta = 0.18 \pm 0.12$ (1σ, N = 4). This estimate is statistically consistent with both $\beta = 0$ ($p = 0.14$, one-tailed) and $\beta \sim 0.2$: the Breu et al. dataset does not discriminate between the two predictions. This ambiguity constitutes the primary scientific motivation for the direct Γ-sweep protocol of Sec. VI, which is statistically powered to resolve it.

At the mechanistic level, the present framework advances beyond Breu et al. [8] by (i) deriving the drag law from the Brinkman–Stokes equation with independently measurable material parameters (κ, η, ρ_{air}), and (ii) identifying Fr_g as the dimensionless control parameter, providing a structural prediction for Γ_c rather than an empirical power-law fit. For completeness, Darcy-flow drag models [5] predict $\beta \approx 0.5$ ($\Gamma^* \propto P^{1/2}$), which is also distinguishable from both $\beta = 0$ and $\beta = 0.18$ at $N_{rep} = 20$.

VIII. DISCUSSION AND SCOPE

A. Domain of Applicability

The framework is valid for: spherical grains; $P \in [10, 200]$ kPa; $f \in [20, 200]$ Hz; $\rho^* \in [0.3, 2.0]$. The lower pressure bound follows from condition H5 ($R_p/\delta \blacksquare 3$); at P = 12 kPa, $R_p/\delta = 2.8$ and the Lamb expansion introduces an estimated 13% error. The upper density bound reflects linearization of Hertz contacts (H8), which fails in the jammed regime ($\rho^* > 2$). Extension to non-spherical grains requires a shape-factor correction [14]; liquid-immersed beds require a separate derivation.

B. Open Questions and Future Directions

Three open questions merit attention. First, the Brinkman transfer function $g(\delta/\lambda_B, \phi)$ in Eq. (III.3) is invoked but not computed analytically here; its closed form for an oscillating sphere in a Brinkman medium is available in [10, 11] and would sharpen the pressure-exponent prediction. Second, the predicted $\Gamma_c = 6.35$ extrapolates from data at $\Gamma = 3$; direct measurement on an independent system is required. Third, the packing fraction ϕ is treated as constant, whereas compaction dynamics couple ϕ to Γ in vibrated beds [16]; the in-situ measurement requirement of Sec. VI directly addresses this.

IX. CONCLUSION

We have derived a granular Froude criterion, $Fr_g = 1$, for the BNE–IBNE transition in vertically vibrated granular beds under variable interstitial gas pressure. The criterion emerges analytically from the condition for the disappearance of the Lorentzian velocity minimum in reduced-density space, within a Brinkman-corrected oscillatory drag framework that correctly captures the three operative flow regimes across the experimental pressure range.

Three technical contributions distinguish this work from previous approaches. First, the Brinkman–Stokes formulation replaces the free-fluid BBO equation, consistently describing the coupled Stokes–Darcy regime that prevails in the Möbius pressure range and provides the physical basis for the drag law $k_m(P) \propto P^{\alpha(P)}$. Second, the local effective exponent $\alpha_{eff}(P)$ is computed analytically from the Lamb expansion, varying from 0.70 at 101 kPa to 0.49 at 12 kPa, resolving the quantitative discrepancy between the asymptotic $P^{1/2}$ prediction and the empirically fitted $n \approx 0.44$. Third, full uncertainty propagation yields $\Gamma_c = 6.35 \pm 0.72$ (k = 2) and establishes that the appropriate falsification threshold is ±15%, larger than the ±10% criterion adopted in earlier formulations.

The central contribution of this paper is a falsifiable, mechanistically grounded prediction: a direct Γ-sweep at multiple interstitial pressures, at fixed (ϕ, f) on an independent granular system, should yield Γ_c invariant to within

±10%. The proposed experimental protocol — N_{rep} = 20, three pressure decades, in-situ ϕ measurement — achieves power > 0.85 to detect a 10% violation if one exists. Confirmation of pressure-independence on an independent system would constitute the first direct experimental evidence for a granular Froude criterion governing the BNE–IBNE transition — a result of direct relevance to the granular physics community and to the broader study of driven particulate media in confined geometries. Falsification — a shift $|\Delta\Gamma_c|/\Gamma_c > 15\%$ — would definitively rule out any pressure-independent transition condition and direct attention toward mechanisms — Darcy flow, effective-temperature coupling, or higher-order Brinkman corrections — in which Γ_c depends on P. Either experimental outcome represents a decisive advance in the quantitative understanding of pressure-mediated granular segregation.

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Supplemental Material contains: (i) regression tables for B and k^2_{20} with digitized data, fit residuals, and confidence intervals; (ii) Kozeny–Carman permeability calculation; (iii) Womersley number evaluation at each experimental pressure; (iv) power analysis spreadsheet for the falsification protocol of Sec. VI.